



School: Campus:

Academic Year: Subject Name: Subject Code:

Semester: Program: Branch: Specialization:

Date:

Applied and Action Learning

(Learning by Doing and Discovery)

Name of the Experiment : Security First – Understanding Blockchain

* Coding Phase: Pseudo Code / Flow Chart / Algorithm

- **Establish the Blockchain Environment:**
Configure the decentralized network by setting up nodes, validators, and secure communication links among participants.
- **Observe Network Operations:**
Monitor the transaction flow — from broadcasting and verification to block creation and addition to the ledger.
- **Simulate Potential Weaknesses:**
Introduce controlled vulnerabilities such as excessive node authority, creation of fake identities, or transaction delays.
- **Initiate Attack Scenario:**
Conduct simulated attacks like the 51% attack or double-spending to test system resilience.
- **Assess System Behavior:**
Record and analyze how the blockchain reacts — identifying issues like detection speed, consensus disruption, or data inconsistency.
- **Implement Security Enhancements:**
Apply mitigation techniques such as stronger consensus algorithms, multi-node authentication, and advanced encryption standards.
- **Verify Network Restoration:**
Evaluate the system's recovery to confirm that normal, secure, and trustworthy operations are re-established after the defense mechanisms are applied.

Software used:

1. VS Code.
2. MS Word.
3. Brave for researching.

* Implementation Phase: Final Output (no error)

- Blockchain network is initialized with multiple nodes.
- Normal transaction flow and block creation are observed.
- A specific attack scenario (e.g., Sybil or 51% attack) is introduced.
- Network disruption or delayed confirmation is noticed.
- The system applies preventive measures (e.g., stake limits, identity verification).
- Blockchain resumes stable operation with secure consensus restored.
- Final output confirms that **security protocols successfully defend against threats**.

* Observations:

- The security of a blockchain network largely relies on the reliability of its consensus mechanism and the trustworthiness of participating nodes.
- Cyberattacks usually take advantage of weaknesses such as excessive network control, programming flaws, or user-related mistakes.
- Applying layered verification processes and conducting regular audits of smart contracts help minimize potential security risks.
- In certain scenarios, Proof of Stake (PoS) and Proof of Authority (PoA) mechanisms provide stronger defense compared to the conventional Proof of Work (PoW) model.
- Ongoing system monitoring and periodic security assessments are vital to safeguard the network against emerging and adaptive threats.

ASSESSMENT

Rubrics	Full Mark	Marks Obtained	Remarks
Concept	10		
Planning and Execution/ Practical Simulation/ Programming	10		
Result and Interpretation	10		
Record of Applied and Action Learning	10		
Viva	10		
Total	50		

Signature of the Student:

Name :

Regn. No. :

Signature of the Faculty: